

Global Minimum Variance

Manuel Vilchis Rangel
manuelvilchisrangel@gmail.com

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1 Calculation of the Global Minimum Variance equation using Lagrange multipliers

1.1 Equation to minimize and constraint

Starting from our covariance equation for a portfolio of n assets

$$f(w) = w^T \Sigma w$$

Subject to the restriction that the sum of all the weights w_i inside the portfolio should be equal to 1.

$$g(w) = 1^T w = 1$$

1.2 Construction of the Lagrange multiplier

$$\mathcal{L}(x, y, \lambda) = f(x, y) + \lambda(g(x, y) - 1)$$

$$\mathcal{L}(w, \lambda) = w^T \Sigma w - \lambda(1^T w - 1)$$

1.3 Matrix derivation and set to 0

$$\frac{\partial \mathcal{L}}{\partial w} = 2\Sigma w - \lambda 1 = 0$$

$$\frac{\partial \mathcal{L}}{\partial \lambda} = 1 - 1^T w = 0$$

1.4 Isolate the unknown vector

$$\begin{aligned}
 2\Sigma w - \lambda 1 &= 0 \\
 2\Sigma w &= \lambda 1 \\
 \Sigma w &= \gamma 1 \\
 w &= \gamma \Sigma^{-1} 1
 \end{aligned} \tag{1}$$

1.5 Replace in the constraint

$$\begin{aligned}
 1^\top w - 1 &= 0 \\
 1^\top w &= 1 \\
 1^\top (\gamma \Sigma^{-1} 1) &= 1 \\
 \gamma (1^\top \Sigma^{-1} 1) &= 1 \\
 \gamma &= \frac{1}{1^\top \Sigma^{-1} 1}
 \end{aligned} \tag{2}$$

1.6 Replace γ in w

$$\begin{aligned}
 w &= \gamma \Sigma^{-1} 1 \\
 w &= \frac{\Sigma^{-1} 1}{1^\top \Sigma^{-1} 1}
 \end{aligned} \tag{3}$$

2 What would happen if Σ were nearly singular?

The problem arises when inverting the covariance matrix, recalling the formula for the inverse of a matrix:

$$\Sigma^{-1} = \frac{1}{\det(\Sigma)} \cdot \text{adj}(\Sigma)$$

A matrix is singular if its determinant is 0, therefore, if it is nearly singular, dividing each element of the adjugate matrix by a value close to 0 causes those elements to explode toward infinity.

When these unstable values are plugged into the GMV formula, the large numbers scale up both the numerator and denominator.

While the final weights are successfully normalized to sum to 1, this amplification forces the portfolio into extreme long and short positions that reflect noise and collinearity, rather than the true underlying relationship between the assets.